



MULTI ISTA

GHG Inventory Report

Apr 2024– Mar 2025

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1 FOREWORD: CARBON FOOTPRINT & GHG PROTOCOL

1.1 CARBON FOOTPRINT

The carbon footprint is a parameter that represents the total emissions of CO₂ and other greenhouse gases (GHG), expressed in mass of CO₂ equivalent, caused directly or indirectly by a product, organisation, service, or event throughout its life cycle. The carbon footprint is important to try to quantify the main emission sources and to have a complete picture of the impact of an organisation on climate change. It is also the first step to carry out a plan to reduce GHG emissions.

The carbon footprint of an organisation intends to quantify the GHG emissions implied by the activity flows of an interconnected entity or group of entities, which may be under its responsibility or on which it depends, over a period of one year with an expressed result in tonnes of CO₂ equivalent (CO₂e).

The calculation of the carbon footprint is more than GHG emissions data, it allows to identify the main GHG emission sources of an organisation and to have a global image of its impact on climate change. Furthermore, it constitutes a necessary base to address and continue over time actions to reduce this impact. Therefore, although the calculation of the carbon footprint by an organisation is voluntary, its assessment has an important strategic aspect and involves a large number of environmental, economic, and reputational benefits:



Knowledge about the environmental impact of an organisation and its contribution to climate change is enriched.



It allows to know and identify the energy consumption and the main GHG emission sources of an organisation, which is a point of reference to design strategies aimed at a better management of the energy used and to prioritise reduction actions with the application of more efficient techniques.



It allows to identify the company's activities with a greater potential for reducing GHG emissions and to set specific objectives for them.



It facilitates the assessment of the choice of raw materials, selection of suppliers, manufacturing methods and production options according to their associated GHG emissions.



It favours the application of more efficient techniques in different activities, thus assuming cost savings.



It is an advance to future regulations and policies on climate change. A clear example is that the EU is already working on how to introduce the calculation of the carbon footprint in the green public procurement.



It means more transparent communication about the company's commitments to sustainable development and, more specifically, the reduction of GHG.



To achieve these objectives, it is necessary to work with the greatest accuracy, covering the maximum possible quantity of emissions for which the organisation is responsible.

Conducting an inventory of greenhouse gas emissions is an important first step a company can take towards developing an effective response to climate change. A greenhouse gas inventory provides valuable information on the risks and opportunities of operating in a carbon constrained economy. At a strategic level, greenhouse gas emissions may be relevant to a company's "license to operate", competitive environment, "carbon risk", and issues of corporate social responsibility. At an operational level, greenhouse gas emissions information may be relevant to decisions on what products to make, what materials and technologies to use, and from where to source energy.

The Greenhouse Gas Protocol, a broad collaboration of businesses, NGO's, governments, and others, was convened by the World Business Council for Sustainable Development (WBCSD) and World Resources Institute (WRI). It represents a voluntary international standard for accounting and reporting greenhouse gas emissions that will enable businesses to report information from global operations in a way that

presents a clear picture of GHG risks and reduction opportunities, while facilitating understanding and comparison with similar reports.

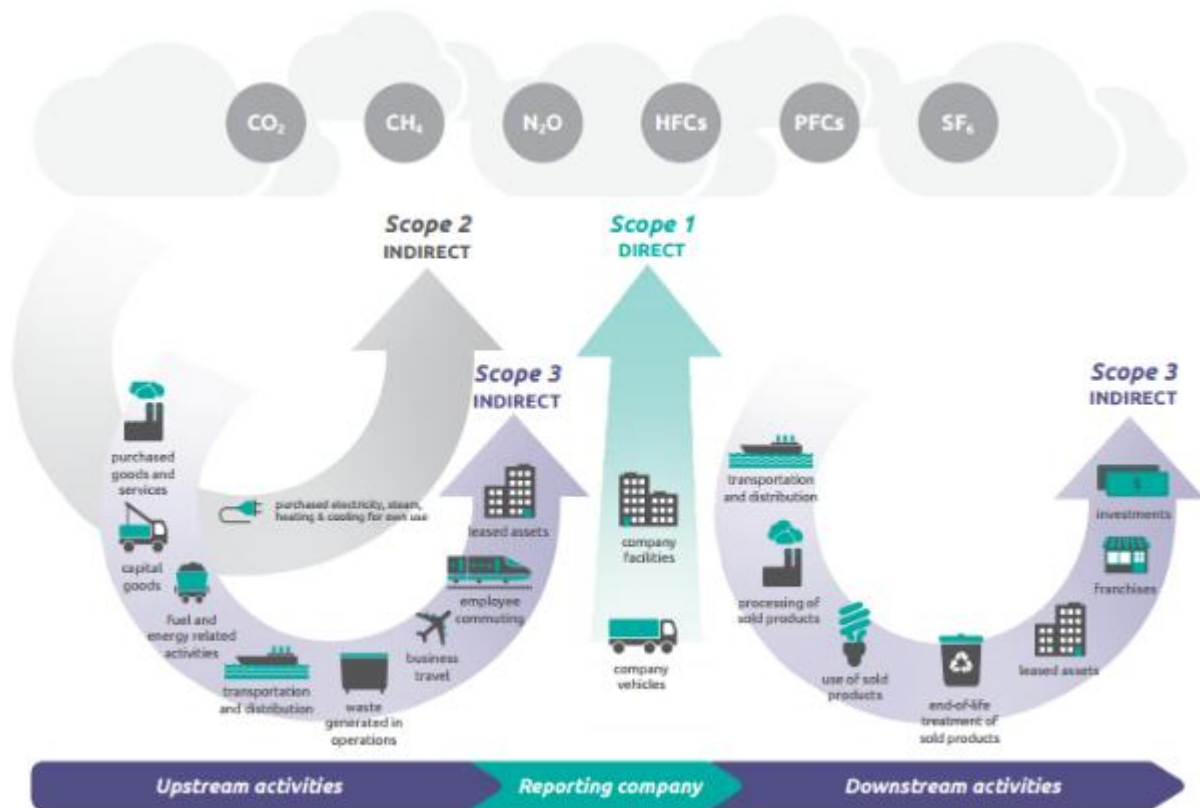
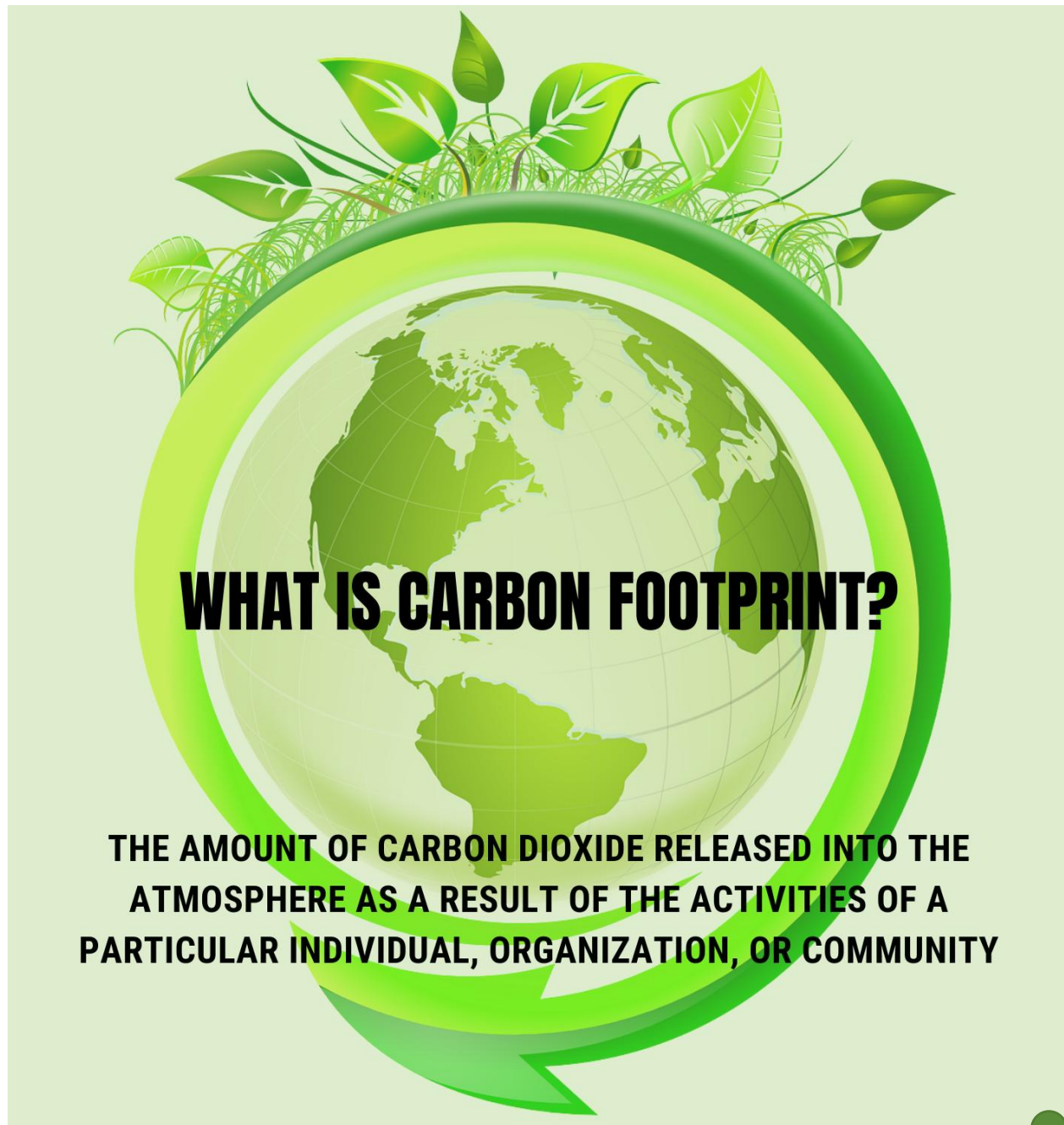


Figure 1: GHG Emissions



The Greenhouse Gas Protocol, a broad collaboration of businesses, NGO's, governments, and others, was convened by the World Business Council for Sustainable Development (WBCSD) and World Resources Institute (WRI). It represents a voluntary international standard for accounting and reporting greenhouse gas emissions that will enable businesses to report information from global operations in a way that presents a clear picture of GHG risks and reduction opportunities, while facilitating understanding and comparison with similar reports.

1.2 GHG PROTOCOL

The Greenhouse Gas Protocol has established a set of guidelines to adhere to in the pursuit of mitigating greenhouse gas (GHG) emissions. The guidelines developed by this organisation are widely utilised globally for carbon accounting purposes in various frameworks such as CDP and other Environmental, Social, and Governance (ESG) frameworks.

The GHG protocol was formulated in the late 1990s by the World Resources Institute (WRI) and the World Business Council for Sustainable Development (WBCSD). Its inception coincided with the initial stages of global endeavours to tackle climate change. Currently, it provides a set of broadly acknowledged criteria for quantifying and disclosing levels of greenhouse gas emissions.

The standards of this framework are employed by governmental bodies, municipalities, corporations, and institutions to convert their operations into measurable inventories of carbon dioxide and other emissions data.

GHG Protocol Corporate Standard establishes a process for quantifying GHG emissions for the inventory. The first steps of this process are identification of specific emission sources within the operational boundaries as well as selection of an emissions quantification methodology applicable for the sources identified.

The next steps are the collection of data required by the methodology for the source and the identification of established emission factors for the data collected. Finally, the data and the emission factors, applied consistent with the quantification methodology, are used to quantify emissions from individual emission sources. The emissions quantified for each source are then consolidated with the other sources within the operational boundaries but ensuring that direct and indirect sources are kept separate.

The Greenhouse Gas Protocol is the main global standard for public and private sector entities to measure emissions. Its standards apply to operations, value chains, and climate change mitigation actions. The purpose of this is twofold: It helps us track and monitor emissions for individual entities and supports greenhouse gas reductions by helping companies identify the most effective ways to reduce their climate impact.

The GHG Corporate Standard and Value Chain Standards are the two most useful standards corporations can use. The Corporate Standard helps companies measure and keep GHG inventories of their operations, while the Value Chain Standard helps them undergo carbon accounting across their value chain.

The GHG Corporate Accounting and Reporting Standard provides the guidance for GHG accounting principles, inventory scopes, GHG emissions sources, setting a base year, and periodically tracking and monitoring emissions. This report is prepared in accordance with GHG Corporate Accounting & Reporting standard.

2 INTRODUCTION

2.1 ABOUT THE REPORT

This Carbon footprint assessment report describes Multivista Global Print Solutions – MVGP’s GHG emissions inventory in the financial year 2024-2025 covering its operations in India which is created in accordance with the GHG protocol corporate reporting standard. The report aims to improve the sustainability performance of MVGP by representing an accurate evaluation of the GHG emissions related to organization’s activities and facilities. Evaluating principle GHG emissions sources will further aid in finding ways to reduce them significantly

2.2 REPORTING PERIOD (BASE YEAR AND REPORTING YEAR)

The GHG emissions inventory described in this report covers MVGP’s GHG Emissions for the base year FY 2019, FY 2022, FY 2023 and current reporting year FY 2024.

2.3 BASE YEAR GHG INVENTORY

The baseline GHG inventory is considered as the previous reporting year FY 2019.

In case of any changes to the operational boundary or change in ownership and control of GHG sources or sinks (transferred into or out of the organizational boundary) the base year GHG emissions and GHG removals will be recalculated based on the methodologies and tools used in preparing this GHG Inventory.

Furthermore, if there are any changes to quantification methodologies that results in significant changes to quantified GHG emissions or removals the base year GHG Inventory will be calculated based on the latest revised version of the calculation methodology.

2.4 ABOUT THE ORGANIZATION

Multivista Global Print Solutions is a leading Printing house in India, equipped with latest, state of the art Printing and Binding machines. They specialize in the art of printing and manufacturing educational books

Started in 1974 as a modest letterpress unit, over the years, they have grown into a fully integrated print house with a host of single, two and four color machines. Currently Multivista boasts of highly sophisticated Heidelberg and Ryobi printing machines and Muller Martini and Kolbus online binding machines.

Over the years they have become the preferred print house for various domestic and international clients primarily because:

- Multivista believes in investing in state of the art machinery.
- Multivista sources reliable and high quality raw materials.
- Multivista employs skilled, efficient and technically qualified workforce.

This has not only improved their reliability but has also enhanced their capability to produce high quality books.

Multivista Global Print House is located in Chennai very close to a Sea and Airport. This not only gives the publisher the flexibility and choice for the mode of transport but also translates into quicker delivery of goods to destinations and is economical in terms of freight costs.

Their success is attributed mainly to the flexibility with which they adapt to different kinds of jobs. They specialize in the niche market of educational books and cater to large publications. Multivista has committed to developing a science-based emission reduction target within the timeframe and in accordance with the SBTi commitment's criteria by signing it as the first Indian Printing Press to do so in 2021.

3 CARBON ACCOUNTING

Carbon Accounting can be considered as the key tool in the efforts to combat climate change. It not only provides ways to quantify carbon emissions but also helps us take informed decisions with respect to GHG mitigation strategies. In order to measure carbon emissions from its business activities, organizations calculate its carbon footprint annually or create quantified GHG inventory.

Each organization's GHG emissions profile encompasses a range of emission sources, each having different levels of impact or Global Warming Potential (GWP).

Carbon Accounting can be categorized into two types:

-  **Physical Carbon Accounting**
-  **Financial Carbon Accounting**

Physical Carbon Accounting is also called GHG inventory. It allows companies to calculate its GHG emissions that they are emitting into the atmosphere. Whereas financial carbon accounting gives a market value for carbon.

3.1 METHODOLOGY USED

IPCC and GHG protocol are some of the common methodologies used for calculating carbon emissions. GHG protocol provides calculation tools for accurate measurements of carbon emissions. Hence, this report is made in accordance with GHG protocol Corporate Accounting and Reporting standard for calculations. The methodology can be summarized as follows:

-  Identification of emission sources
-  Collection of GHG activity data
-  Selection of emissions factors
-  Selection of quantification methodology
-  Calculation of Greenhouse gas emissions

3.2 CARBON ACCOUNTING STANDARDS

Most widely recognized international standards for carbon accounting is GHG protocol, ISO 14064-1:2018. The ISO standard framework helps organizations by providing a methodology on how to account emissions and identify carbon baseline values. For developing a GHG inventory for organizations, it requires three important aspects such as setting inventory boundaries, quantifying, and reporting GHGs.

GHG Corporate Standard guides organizations in identifying specific emissions sources within the operational boundaries as well as provides methodologies to quantify those emissions.

4 BOUNDARIES

4.1 ORGANIZATIONAL BOUNDARIES

- **Location:** Multivista Global Print Solutions (MVGP) located at 43, Vandalur Kelambakkam Road, Pudupakkam, Chennai - 603 103

In accordance with the GHG Protocol – Corporate Standard for setting the scope and boundary of the organisation, MVGP’s Pudupakkam site has been considered for carbon accounting. MVGP’s Scope 1 and Scope 2 emissions from all its operations are accounted.

- **Reporting Period:**
 - Base Year: Apr 19 – Mar 2020
 - Current Reporting Year: Apr 2024 – Mar 2025

4.2 OPERATIONAL BOUNDARIES

MVGP has confirmed and documented its operational boundaries and has identified the GHG emissions from its operations and removals associated with the company.

Operational boundaries are defined in three different scopes:

Scope 1 emissions include all direct CO₂ emissions emitted from its operational activities. These emissions include sources owned or controlled by MVGP such as energy and fuel consumption and other emissions in the form of refrigerant.

Scope 2 emissions include energy indirect emissions emitted from the consumption of purchased electricity by MVGP. This emission occurs at the source where electricity is produced.

Scope 3 emissions includes emissions from purchased goods, capital equipment, business travel, employee commute, upstream and downstream transportation and distribution.

5 DATA COLLECTION AND QUANTIFICATION METHODOLOGIES

5.1 COLLECTION AND MONITORING METHODOLOGY

Activity data is collected at the site level by the facility management team and consolidated by management team. Although currently there is no automated data collection system in place, data is accurately collected every month for better management of scope wise emissions.

MVGP facility management constantly explores several ways to collect and aggregate data using existing processes and infrastructure. For instance, energy consumption data is monitored and collected manually to ensure there is no leakage.

Below are the sources used to collect data

-  Interval meter information
-  Utility invoices (at Group level, sub meter information is used for MVGP information)
-  Bills (Electricity, Diesel, Fuel)

5.2 GHG EMISSION ACTIVITY DATA SOURCES

Table 1: GHG Emission Activity Data Sources

Description	GHG Activity Data is based on
Direct GHG Emissions – Diesel Consumption for Diesel for DG, CO ₂ consumption for Fire extinguishers.	Bills/Invoice/Meter Information
Indirect GHG Emissions – Purchased Electricity	Bills/Invoice/Meter Information as applicable
Scope 3 emissions - Emissions from purchased goods and services, capital goods, upstream and downstream transportation, waste disposal and business commute, employee commute	Bills/Invoice.

5.3 GHG SOURCES AND GHG INFORMATION MANAGEMENT TEAM

There is a team appointed by management for maintaining data and corresponding evidence for GHG emission sources. Mainly Environmental, Health & Safety, Utility & admin team is responsible for data collection in the organization with help of support functions for data collection and validation.

5.4 ASSUMPTIONS

For all emission sources an emission factors were identified as per India GHG Protocol & Defra 2023 emission factors, USEEIO database and CEA database.

5.5 FREQUENCY OF REPORT

The GHG Inventory report is prepared annually since the base reporting year FY 2019.

5.6 UNCERTAINTY OF ASSESSMENT

The data for GHG emissions from MVGP has been derived directly from electricity, logbook/invoices/meter information of DG & Electricity Board, purchase records, commute bills/registers/invoices and MIS records and then conversion factors are employed to convert into GHG reporting metrics. The uncertainty lies in the conversion factors.

6 RESULTS

6.1 PRINCIPLES OF GHG ACCOUNTING

GHG accounting and reporting practices are continuously evolving along with advancing knowledge on the science of climate change. The GHG protocol standard advise that GHG emissions inventories be carried out in accordance with the following principles:

-  **RELEVANCE-** For an organization's GHG emissions inventory to contain information that users might need for making "informed" decisions. Accordingly, MVGP has identified the appropriate boundaries that reflects their business operations.

- 
COMPLETENESS - All relevant emission sources within the chosen inventory boundary have been accounted in the GHG inventory so that a comprehensive and meaningful inventory of total emissions are compiled.
- 
CONSISTENCY - The GHG inventory has been compiled in a manner that ensures that the overall emissions estimate is consistent and comparable over time.
- 
TRANSPARENCY –All necessary information has been recorded, compiled, and analysed in a manner that enables internal reviewers and external verifiers to attest to its credibility.
- 
ACCURACY - Data reported is sufficiently precise to enable us to make decisions with reasonable assurance and the reported information is credible. Uncertainties in measurements, recording and calculations has been reduced as far as possible and practicable.

6.2 TOTAL GHG EMISSIONS IN APRIL 2019– MARCH 2020 (BASE YEAR)

Table 2: Total GHG Emissions April 2019 – March 2020

Emission Category	GHG Emissions in mtCO ₂ e
Scope 1	255
Scope 2	1502
Total Emissions	1756

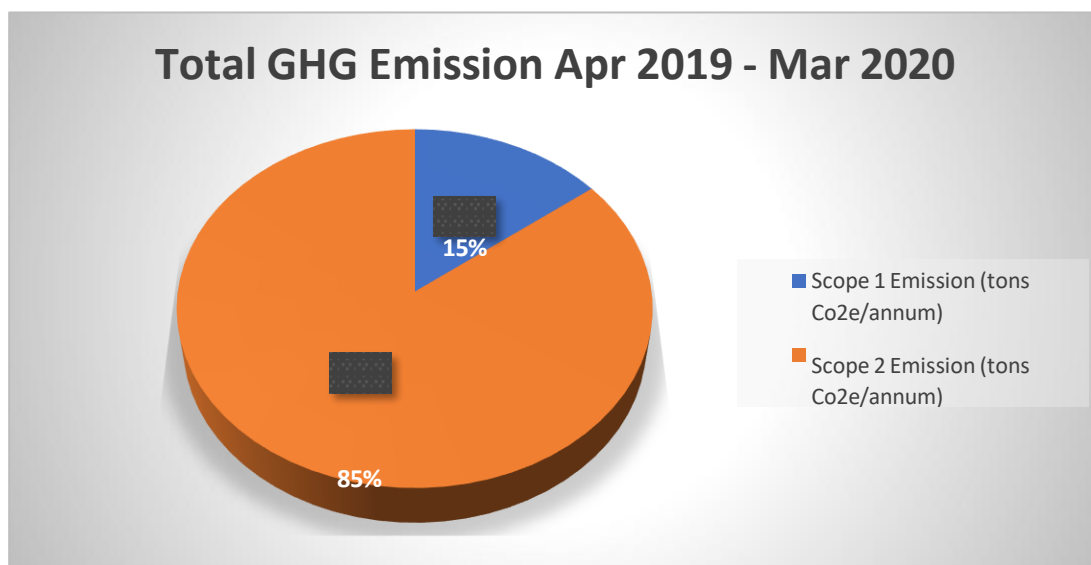


Figure 2: Total Emissions for base year April 2019 – March 2020

Scope 2 emissions contribute 85 % of the total emissions for MVGP.

6.3 TOTAL GHG EMISSIONS IN APRIL 2023 – MARCH 2024

Table 3: Total GHG Emissions April 2023 – March 2024

Emission Category	GHG Emissions in mtCO ₂ e
Scope 1	276
Scope 2	142.85
Scope 3	4268.47
Total Emissions	4687.31

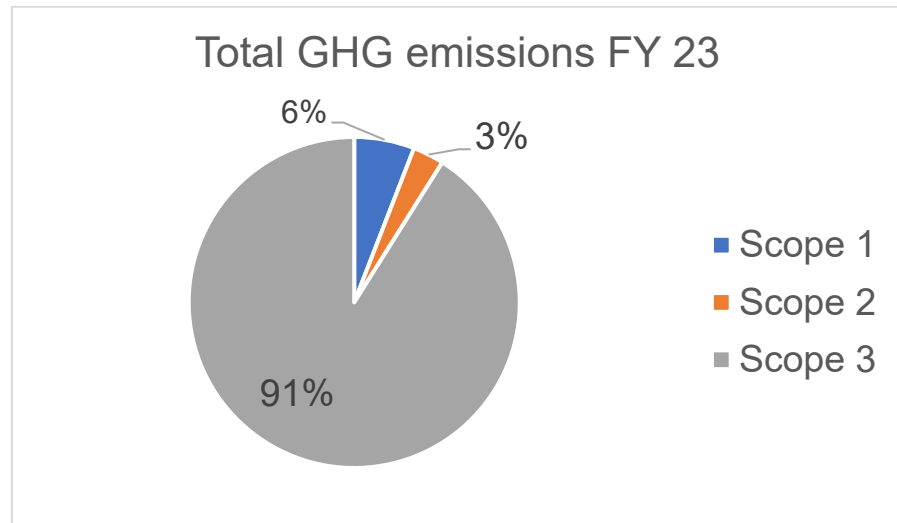


Figure 3 :Total Emissions for FY year April 2023 – March 2024

Scope 3 emissions contribute 91 % of the total emissions for MVGP.

6.4 SCOPE 1 EMISSIONS: DIRECT GHG EMISSIONS

Direct GHG emissions occur from sources that are owned or controlled by the company, for example, emissions from Refrigerant & Fire extinguisher

Following is the chart that shows various emission sources of scope 1 in 2023. Among all areas, emissions from split AC system has the highest emission.

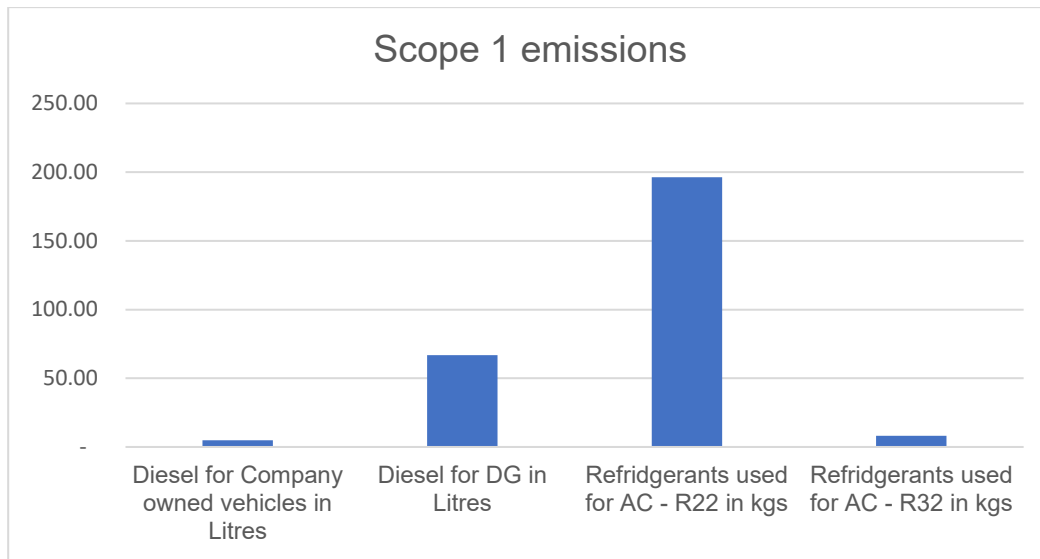


Figure 4: Scope 1 emissions by source in mtCO2e in FY 23

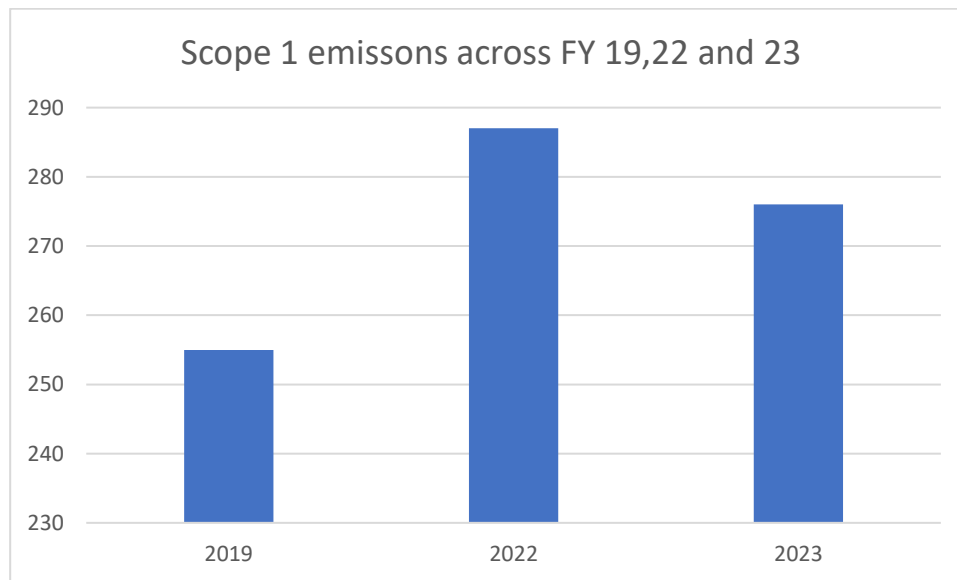


Figure 5: Comparison of Scope 1 emissions across the FY 19,22 and 23

Scope 1 emissions for a MVGP increased from 255 units in 2019 to 287 units in 2022, before slightly decreasing to 276.001 units in 2023. This indicates a significant rise in emissions from 2019 to 2022, followed by a modest improvement. To further reduce emissions, the company could invest in energy-efficient equipment, switch to renewable energy sources, optimize processes, and enhance employee training on energy-saving practices.

6.5 SCOPE 2: ELECTRICITY INDIRECT GHG EMISSIONS

Scope 2 accounts for GHG emissions from the purchased electricity consumed by a company. Purchased electricity is defined as electricity that is purchased or otherwise brought into the organizational boundary of the company. Scope 2 emissions physically occurs at the facility where electricity is generated.

For the year 2023, Scope 2 emissions of MVGP is 142 mt CO₂e. Following chart shows the split of various energy sources of MVGP and only non-renewable electricity sources are considered for the calculations.

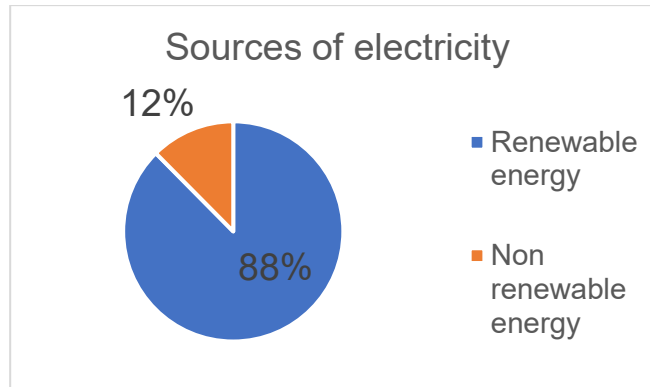


Figure 6: Sources of electricity for MVGP in 2023

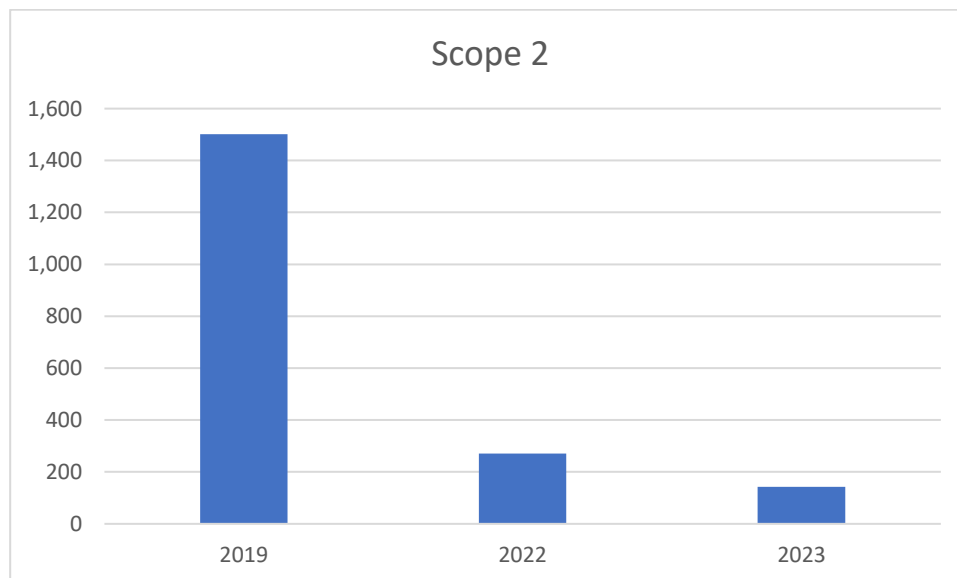


Figure 7: Comparison of Scope 2 emissions across FY19, FY22 and FY23

Scope 2 emissions show a significant reduction, from 1502 units in 2019 to 271 units in 2022, and further down to 142.88 units in 2023. This indicates an impressive decrease of about 82% from 2019 to 2022, followed by an additional 47% reduction in the following year. These reductions likely result from measures

such as switching to renewable energy sources, improving energy efficiency, and optimizing electricity consumption. Continued efforts in these areas can help the company maintain and further this positive trend in reducing Scope 2 emissions.

6.6 SCOPE 3 EMISSIONS FOR FY 23

Scope 3 emissions encompass all indirect emissions that occur in a company's value chain, excluding those covered in Scope 2. This includes emissions from activities such as procurement, transportation, waste management, and the use of sold products.

MVGP's Scope 3 emissions, totaling 4268.47 units, are being reported for the first time this year, marking a significant step in their environmental accountability. The largest contributor is Category 1 (Purchased Goods) at 3611.40 units, highlighting the substantial impact of material procurement on their carbon footprint. Emissions from Upstream and Downstream Transportation (356.32 units) and Employee Commutation (152.61 units) also play notable roles, emphasizing the environmental costs of logistics and daily travel. Additionally, Business Travel (65.56 units) and Waste Management (12.75 units) contribute to the overall emissions. The company recognizes the importance of these findings and is committed to implementing strategies to reduce their Scope 3 emissions in the coming years. The following chart illustrates the various sources of Scope 3 emissions for MVGP,

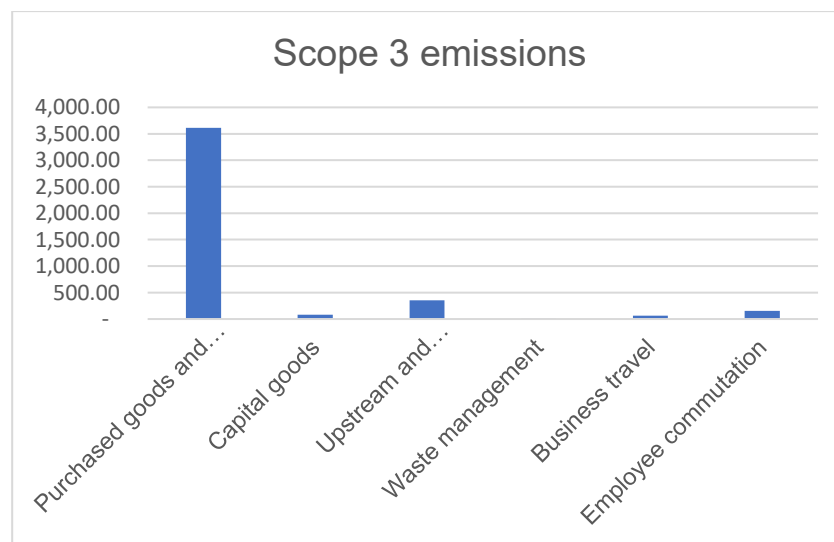
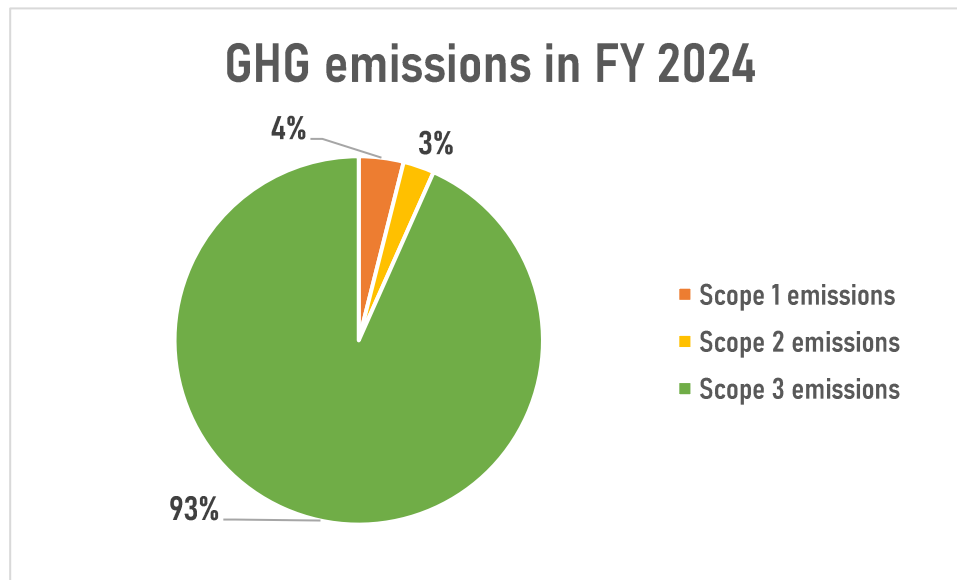


Figure 8: Sources of Scope 3 emissions for FY 23

6.7 TOTAL GHG EMISSIONS IN FY 2024 (CURRENT REPORTING YEAR)

Emission Category	GHG Emissions in mtCO ₂ e
Scope 1	206.33
Scope 2	143.462
Scope 3	4900.54
Total Emissions	5250.297

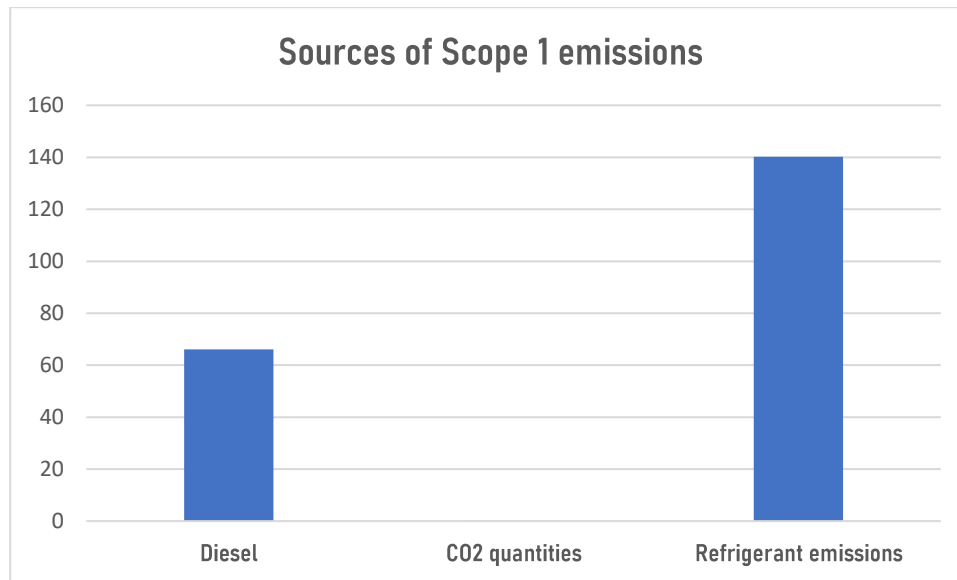


6.8 SCOPE 1 EMISSIONS: DIRECT GHG EMISSIONS

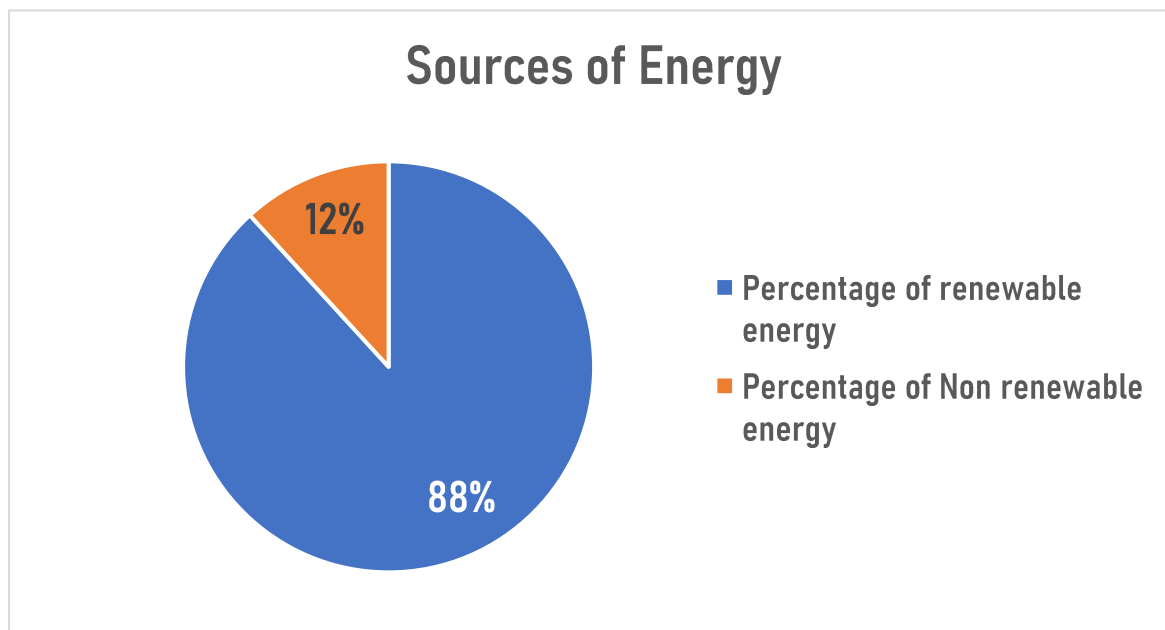
Direct GHG emissions occur from sources that are owned or controlled by the company, for example, emissions from refrigerant usage, diesel combustion, and CO₂-based fire suppression systems.

Following is the chart that shows various emission sources of Scope 1 in FY 2024. Among all sources, emissions from refrigerants—particularly from R22 used in split AC systems—accounted for the highest share of emissions. Diesel consumption from backup generators was the second-largest contributor, followed by emissions from R410A, R32, and minor releases of CO₂ from extinguishers or operational activities.

To mitigate Scope 1 emissions in the coming years, MVGL plans to transition away from high-GWP refrigerants like R22 and R410A by replacing them with low-GWP alternatives such as R32, R290, or HFO-based systems. Additionally, regular maintenance of HVAC systems, installation of leak detectors, and training of technical staff on refrigerant handling will be prioritized. Diesel emissions will be addressed through increased use of biodiesel blends, on-site solar power systems, and electrification of small operational equipment to reduce fossil fuel reliance.



6.9 SCOPE 2 EMISSIONS: ELECTRICITY INDIRECT GHG EMISSIONS



Indirect GHG emissions from Scope 2 occur from the generation of purchased electricity consumed by the company. For MVGL, these emissions are associated with the power bought from the local utility grid to run its operational facilities, lighting, and equipment.

The chart below illustrates MVGL's overall energy consumption mix for FY 2024. All Scope 2 emissions, totaling 143.46 tCO₂e, are attributable to a single source: purchased grid electricity. This component, while essential for continuous operations, represents the 12% of MVGL's total energy consumption that is non-

renewable. The vast majority of the company's energy profile (88%) is dominated by self-generated renewable wind energy, which does not contribute to Scope 2 emissions.

To mitigate Scope 2 emissions in the coming years, MVGL will focus on reducing its reliance on purchased grid electricity. Key strategies include implementing comprehensive energy efficiency measures, such as upgrading to LED lighting, installing energy-efficient machinery, and optimizing building management systems. The company will also explore options for procuring certified renewable energy through Power Purchase Agreements (PPAs) to address any remaining electricity demand.

6.10 SCOPE 3 EMISSIONS FOR FY 24

Scope 3 emissions encompass all indirect emissions that occur in a company's value chain, excluding those covered in Scope 2. This includes emissions from activities such as procurement, transportation, waste management, employee commuting, and downstream product use. Multivista's Scope 3 emissions, totaling **4900.50 metric tonnes of CO₂e**, reflect a comprehensive understanding of its extended environmental impact. The largest contributor is Category 1 (Purchased Goods and Services) at **4030.64 tCO₂e**, underlining the dominant influence of procurement in the company's carbon footprint. Emissions from Capital Goods (234.85 tCO₂e) and Downstream Activities (353.90 tCO₂e) also represent significant contributions.

Other notable emission sources include Employee Commute (167.13 tCO₂e), Business Travel (50.18 tCO₂e), and Fuel-Related Activities (19.10 tCO₂e), highlighting the impact of daily operations and travel on sustainability performance. Upstream Transportation (39.36 tCO₂e) and Waste Disposal (5.33 tCO₂e) contribute smaller shares but are still important elements of the full emissions picture.

The company acknowledges the importance of understanding and managing Scope 3 emissions and is committed to identifying opportunities for reduction and collaboration across its value chain. The following chart presents the distribution of Scope 3 emissions across different categories:

6.11 IMPROVEMENT IN FY 2024 CARBON EMISSIONS AGAINST FY 2019

Multivista has demonstrated substantial progress in reducing its greenhouse gas emissions in 2024 compared to the base year of 2019, particularly through strategic efforts to decarbonize its electricity consumption and optimize operational efficiency. The most significant achievement lies in Scope 2 emissions, which have seen a dramatic 90.48% reduction, decreasing from 1,502 tons CO₂e in 2019 to just 143 tons CO₂e in 2024. This improvement reflects a focused transition towards renewable energy and improved energy management practices.

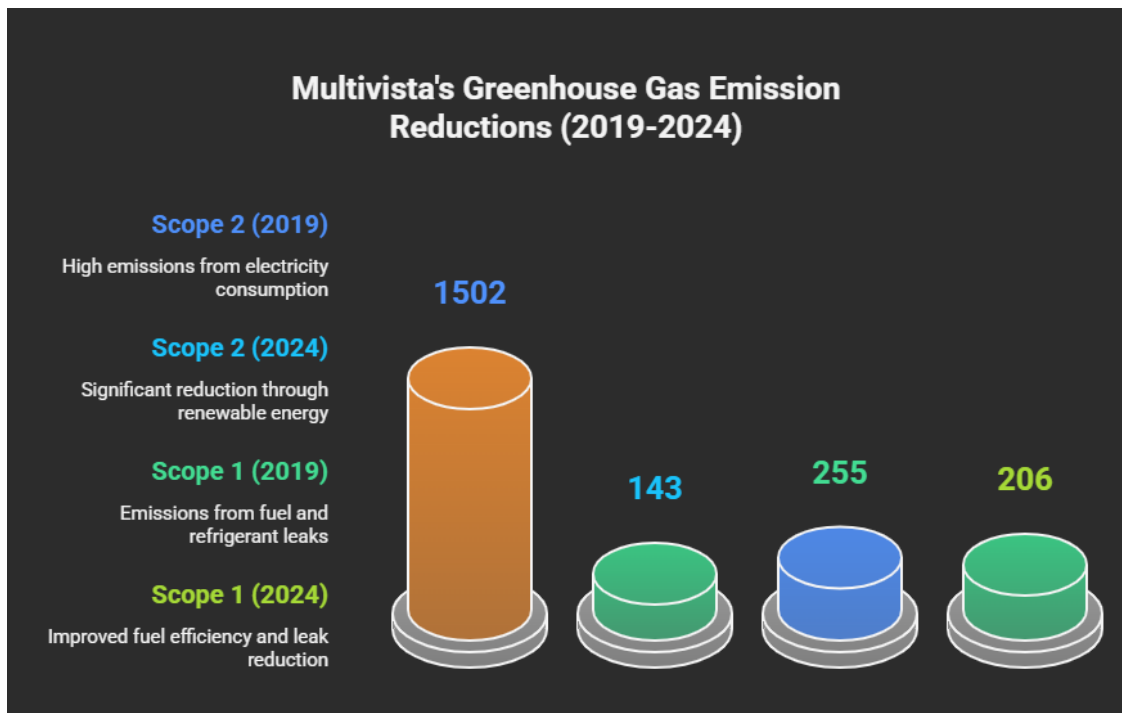
In terms of Scope 1 emissions, Multivista has achieved a 19.22% reduction, bringing emissions down from 255 tons CO₂e in 2019 to 206 tons CO₂e in 2024. This decline indicates improved fuel efficiency and possible reductions in on-site combustion and refrigerant leaks.

These consistent year-on-year reductions underscore Multivista’s commitment to sustainability and climate responsibility. The company’s approach not only emphasizes reducing absolute emissions but also reflects the importance of transitioning to low-carbon operations.

As operational scale and productivity can influence absolute emissions, it is recommended that Multivista complements these figures with intensity-based metrics—such as emissions per ton of production output—to more accurately assess performance improvements and drive further targeted reductions.

Table 3: Comparison and improvement in Carbon emissions (Base year 2019 vs FY 2022 vs Current reporting year 2023)

Year	Scope 1 emissions Tons Co2e	Scope 2 emissions Tons Co2e
2019	255	1,502
2022	287	271
2023	276	142
2024	206	143
Reductions in Emissions from 2019-2024	19.22% reduction achieved.	90.48 % reduction achieved.



Financial Year	2019	2022	2023	2024
Carbon Intensity – kg CO ₂ e/per unit of book produced (Only Scope 1 and scope 2 emissions)	0.56	0.13	0.043	0.042
Carbon intensity – kg CO ₂ e / per unit of book produced(Scope 1,2 and 3 emissions)	-	-	0.42	0.63

6.12 SUMMARY OF GHG PERFORMANCE: FY 2023 vs. FY 2024

Multivista has demonstrated commendable progress in managing its operational greenhouse gas emissions between FY23 and FY24. The company achieved a significant 16.5% reduction in its combined Scope 1 and 2 emissions, which decreased from 418 to 349 tonnes of CO₂e. This positive trend is further reflected in the improved operational efficiency, as the carbon intensity for Scope 1 and 2 emissions per book produced also saw a slight decrease from 0.043 to 0.042 kg CO₂e. These figures underscore the success of internal initiatives aimed at reducing direct emissions and optimizing energy consumption.

Despite these operational gains, a critical challenge is highlighted by the broader carbon intensity metrics. The comprehensive carbon intensity, which includes Scope 3 emissions, increased from 0.42 kg CO₂e/unit in FY23 to 0.63 kg CO₂e/unit in FY24. This increase is primarily attributed to a rise in Scope 3 emissions resulting from significant capital expenditure (capex) during the year. The construction work undertaken to expand the operational setup has led to substantial one-time emissions associated with capital goods, such as building materials and construction activities. While this investment is crucial for future growth, it underscores the importance of expanding focus to the entire value chain. Going forward, embedding sustainable procurement criteria for capital projects and managing other Scope 3 categories will be vital to achieving holistic and long-term carbon reduction.

7 TOWARDS CARBON REDUCTION

A Practical Roadmap for a Leader in Print

Multivista has already made commendable progress in reducing its carbon footprint, particularly in its Scope 2 emissions. This plan outlines the next strategic steps to build on that success, focusing on practical, industry-specific initiatives that will drive further reductions, enhance operational efficiency, and solidify Multivista's position as a sustainable leader in the printing industry.

7.1 OPTIMIZE AND ELECTRIFY FLEET OPERATIONS

(Scope 1: Emission Reduction Strategy)

Your direct emissions from fuel consumption are a key area for targeted reduction.

- **Implement Route Optimization Software:** Utilize technology to plan the most efficient delivery routes, reducing fuel consumption, mileage, and service hours for your vehicle fleet.
- **Transition to Electric Vehicles (EVs):** Develop a phased plan to replace fossil-fuel-powered delivery and sales vehicles with electric alternatives. Prioritize local delivery routes where EVs are most effective.
- **Eco-Driving Training:** Train drivers in fuel-efficient techniques such as smooth acceleration/deceleration, reduced idling time, and proper tire pressure maintenance.

7.2 REDUCE PROCESS-RELATED EMISSIONS

(Scope 1: Emission Reduction Strategy)

A significant portion of a printing company's Scope 1 emissions comes from the materials used in the printing process itself.

- **Switch to Low/No-VOC Inks and Solvents:** Actively transition from traditional petroleum-based inks and chemical solvents to vegetable-based inks and low-Volatile Organic Compound (VOC) or VOC-free cleaning solutions. This directly reduces fugitive GHG emissions and improves workplace air quality.
- **Upgrade Gas-Fired Equipment:** If using natural gas for drying processes (e.g., in heatset web offset presses), evaluate and upgrade to more efficient burners or explore alternative curing technologies.

7.3 AGGRESSIVELY PURSUE ENERGY EFFICIENCY IN PRODUCTION

(Scope 1 and Scope 2: Emission Reduction Strategy)

The high energy demand of printing and finishing equipment offers the single largest opportunity for further Scope 2 reduction.

- **Conduct a Specialized Energy Audit:** Go beyond a general audit. Commission a Level 2 ASHRAE audit focused on high-demand systems specific to a printing facility:
 - **Pressroom Equipment:** Analyze the energy draw of individual presses during make-ready, run-time, and idle states.

- **Compressed Air Systems:** These systems are notoriously inefficient. Conduct an ultrasonic leak detection survey and ensure compressors and air pressure levels are optimized for the actual equipment demand.
- **HVAC Systems:** Ensure your climate control is zoned and programmed efficiently, maintaining the critical temperature and humidity for paper stock without excessive energy use in non-critical areas.
- **Invest in Energy-Efficient Technology:**
 - **LED-UV Curing:** When upgrading presses, prioritize models with LED-UV curing technology. It consumes significantly less energy than traditional mercury-vapor UV or IR drying systems and generates less ambient heat, reducing the HVAC load.
 - **Modern Press Drives and Motors:** Newer presses and binding equipment feature Variable Frequency Drives (VFDs) and high-efficiency motors that drastically cut electricity consumption.
- **Optimize Production Workflow:** Minimize press idle time through better job scheduling and faster make-ready procedures. Power down equipment completely during extended periods of inactivity.

7.4 ENHANCE SUSTAINABLE SOURCING AND CIRCULARITY

(Future-Proofs for Scope 3 and Reduces Waste-Related Emissions)

Extend your environmental commitment by focusing on your primary raw material—paper—and minimizing waste.

- **Prioritize Certified and Recycled Paper Stock:** Mandate the use of paper certified by the Forest Stewardship Council (FSC) or Sustainable Forestry Initiative (SFI). Actively promote and use paper with high post-consumer recycled content to your clients.
- **Implement Advanced Waste Reduction:**
 - **Smart Imposition:** Use software to optimize job layouts on paper sheets, minimizing offcut waste.
 - **Comprehensive Recycling Program:** Expand your recycling efforts beyond paper offcuts. Implement dedicated streams for aluminum printing plates, ink and toner cartridges, chemical containers, and shrink wrap.
- **Engage with Suppliers:** Partner with ink, paper, and chemical suppliers who share your commitment to sustainability and can provide their own carbon footprint data.

7.5 STRATEGIC USE OF RENEWABLE ENERGY AND OFFSETS

(Scope 2 and Final Balancing Strategy)

After maximizing efficiency and reduction measures, address the remaining emissions.

- **On-Site Solar Generation:** Your facility's large, flat roof is a prime asset for a solar panel installation. This provides a long-term, fixed-cost source of clean energy, directly reducing your Scope 2 emissions and offering significant ROI.

- **Power Purchase Agreements (PPA):** If on-site solar is not immediately feasible, enter a direct PPA with a local renewable energy provider to secure a stable supply of clean electricity.
- **Carbon Offsets as a Final Step:** After all practical reductions have been implemented, offset the remaining, unavoidable emissions. To enhance your brand story, invest in high-quality, certified offset projects that resonate with your industry, such as:
 - **Reforestation and Sustainable Forestry Management Projects.**
 - **Projects that support recycling infrastructure in developing communities.**

7.6 ALIGN BUSINESS STRATEGIES WITH SUSTAINABLE DEVELOPMENT GOALS (SDGs)

MVGP can align business strategies with Sustainable Development Goals (SDGs) to create a positive business impact on environmental, social, and economic development.

Some of the recommended SDGs that MVGP can focus on:



Investing in more resilient infrastructure and technological innovation to support sustainability.



Efficient consumption of resources such as energy, water, and waste.



Partnering with vendors, government agencies to implement green projects.



Incorporating renewable energy into organizational energy mix by procuring green energy from independent power producers using PPA (Power Purchase Agreement)



Investments to enhance campus biodiversity.



Mitigation of Climate Change impacts through sustainable business

7.7 OTHER RECOMMENDATIONS

- Implement route optimization software to create the most efficient delivery schedules and reduce fuel consumption.
- Develop a phased plan to transition the company's vehicle fleet to electric vehicles (EVs), prioritizing local delivery routes.
- Train all drivers in eco-driving techniques, including smooth acceleration, reduced idling, and proper tire maintenance.
- Switch from petroleum-based inks and chemical solvents to vegetable-based, low-VOC, or VOC-free alternatives.
- Upgrade any natural gas-fired drying equipment on presses to newer, more energy-efficient models.
- Conduct a specialized energy audit (e.g., Level 2 ASHRAE) focused specifically on pressroom equipment, compressed air systems, and HVAC.
- Perform an ultrasonic leak detection survey for the compressed air system to identify and repair costly air leaks.
- Prioritize investing in energy-efficient press technology, such as LED-UV curing systems, when upgrading equipment.
- Ensure new machinery is equipped with high-efficiency motors and Variable Frequency Drives (VFDs).
- Optimize production workflows and job scheduling to minimize press idle time and reduce energy waste between jobs.
- Mandate the procurement of paper certified by the Forest Stewardship Council (FSC) or Sustainable Forestry Initiative (SFI).
- Actively promote and use paper stock with high post-consumer recycled content in client projects.
- Utilize smart imposition software to optimize job layouts on paper sheets, significantly reducing offcut waste.
- Establish comprehensive recycling programs for all operational waste, including aluminium printing plates, ink cartridges, chemical containers, and shrink wrap.
- Evaluate and install an on-site solar panel system on the facility's roof to generate clean electricity.
- Explore and secure a Power Purchase Agreement (PPA) with an off-site renewable energy provider.
- After implementing all feasible reductions, purchase certified carbon offsets to address any remaining, unavoidable emissions.
- Invest in offset projects that resonate with your industry, such as reforestation or sustainable forestry management.

- Adopt waterless offset printing technology when purchasing new presses to eliminate water and chemical use in the dampening process.
- Install rainwater harvesting systems to capture and use water for non-potable needs like cleaning and irrigation.
- Conduct a facility-wide water audit to identify and fix leaks, and install low-flow fixtures in restrooms and breakrooms.
- Use thermal imaging cameras for proactive maintenance to find and fix energy inefficiencies in electrical panels and motors before they fail.
- Implement an internal "Right-to-Repair" policy for office electronics and non-critical equipment to reduce e-waste.
- Establish a voluntary, employee-led "Green Team" or Sustainability Committee to champion new initiatives.
- Launch gamified "eco-challenges" and reward programs for employees to encourage sustainable habits like carpooling and waste reduction.
- Develop a formal sustainable procurement policy for all office supplies, requiring recycled, refillable, or eco-certified products.
- Prioritize sourcing materials and supplies from local vendors to reduce supply chain transportation emissions.
- Advise clients during the design phase on using materials and bindings that allow the final product to be easily recycled.
- Create a supplier sustainability scorecard to evaluate and favour partners with strong environmental credentials.
- Deploy smart power strips at all workstations to automatically cut power to peripherals and eliminate phantom energy draw.
- Standardize the use of Energy Star certified products for all new office equipment, including computers, printers, and lighting.
- Address Scope 3 emissions by encouraging green travel options for commutes and business trips, such as public transport, EVs, and staying in green hotels.
- Install a Building Management System (BMS) or sustainability software to automate data collection and monitor resource consumption in real-time.
- Pursue "Green Building" certifications like LEED for any new office construction or major facility renovations.
- Apply white reflective coatings or insulation to the facility's roof to decrease heat absorption and lower air conditioning costs.

8 CONCLUSION

MVGL has demonstrated continued and significant progress in reducing its greenhouse gas (GHG) emissions in FY 2024, reinforcing its deep commitment to environmental stewardship. Building on previous successes, the company has maintained its exceptional performance in Scope 2 emissions, which stand at 143 mtCO₂e. This represents a remarkable 90.5% reduction from the 1,502 mtCO₂e recorded in the base year of FY 2019, an achievement primarily sustained by the strategic procurement of renewable energy and ongoing energy efficiency measures.

In a key positive development this year, Scope 1 emissions have also seen a substantial decline, falling to 206 mtCO₂e from a base of 255 mtCO₂e. This marks a 19.2% reduction in direct emissions since 2019, reflecting successful efforts to improve operational efficiencies. Further reductions can be realized by focusing on proactive maintenance of HVAC systems to prevent refrigerant leaks, exploring Low-Global Warming Potential (GWP) refrigerants, and optimizing fuel consumption within the company's vehicle fleet.

As MVGL successfully manages its direct operational footprint, the next critical step is to expand focus to the value chain. While Scope 3 emissions have not yet been quantified, it is understood that they represent the largest portion of the company's total carbon footprint. Activities such as the procurement of paper and inks, product transportation, business travel, and waste management are significant contributors. A primary goal for the upcoming fiscal year will be to establish a comprehensive methodology for measuring and reporting these Scope 3 emissions to inform future reduction strategies.

Overall, MVGL's dedication to sustainability is clearly evidenced by its ambitious and successful emissions reduction efforts across both Scope 1 and Scope 2. The company's ability to drive down emissions in two distinct areas simultaneously showcases a mature and effective environmental strategy. Moving forward, the focus will be on maintaining the excellent results in Scope 2 while continuing to curb Scope 1 emissions and, crucially, embarking on the new challenge of engaging with suppliers and stakeholders to address Scope 3. Through continued innovation, data-driven strategies, and stakeholder collaboration, MVGL is well-positioned to achieve its long-term sustainability goals and lead the industry in climate action.

9 REFERENCES

For the purpose of calculating and reporting GHG emissions, the following standards, protocols, and emissions factors are applicable:

1. Greenhouse Gas Protocol on Corporate Accounting and Reporting by World Resource Institute (WRI) / World Business Council for Sustainable Development (WBCSD)
2. Defra Greenhouse Gas Protocol <https://ghgprotocol.org/Third-Party-Databases/Defra>
3. GHG Protocol https://cea.nic.in/wp-content/uploads/baseline/2020/07/user_guide_ver15.pdf
4. Central Electricity Authority <https://cea.nic.in/?lang=en>

For any further questions/clarifications, please contact pnkrishna@multivistaglobal.com

ANNEXURE 01: LIST OF ABBREVIATIONS

GHG		Green House Gas	
AC	Air Conditioning	Km	Kilo meter
Avg.	Average	KV	Kilo Volt
CEA	Central Electricity Authority	kWp	Kilo Watt peak
CO2e	Carbon dioxide equivalent	LPG	Liquefied Petroleum Gas
EE&EC	Energy Efficiency and Energy Conservation	MT	Metric Tonnes
GCM	Gramm Cubic Meter	MTCE	Metric Tonnes Carbon Equivalent
GWP	Global Warming Potential	MU	Million Units
IPCC	Intergovernmental Panel for Climate Change	MW	Mega Watt